

# NAARI SHOURYA A

AI Media Protection & Rapid Response Platform

Women Safety & Deepfake Detection System

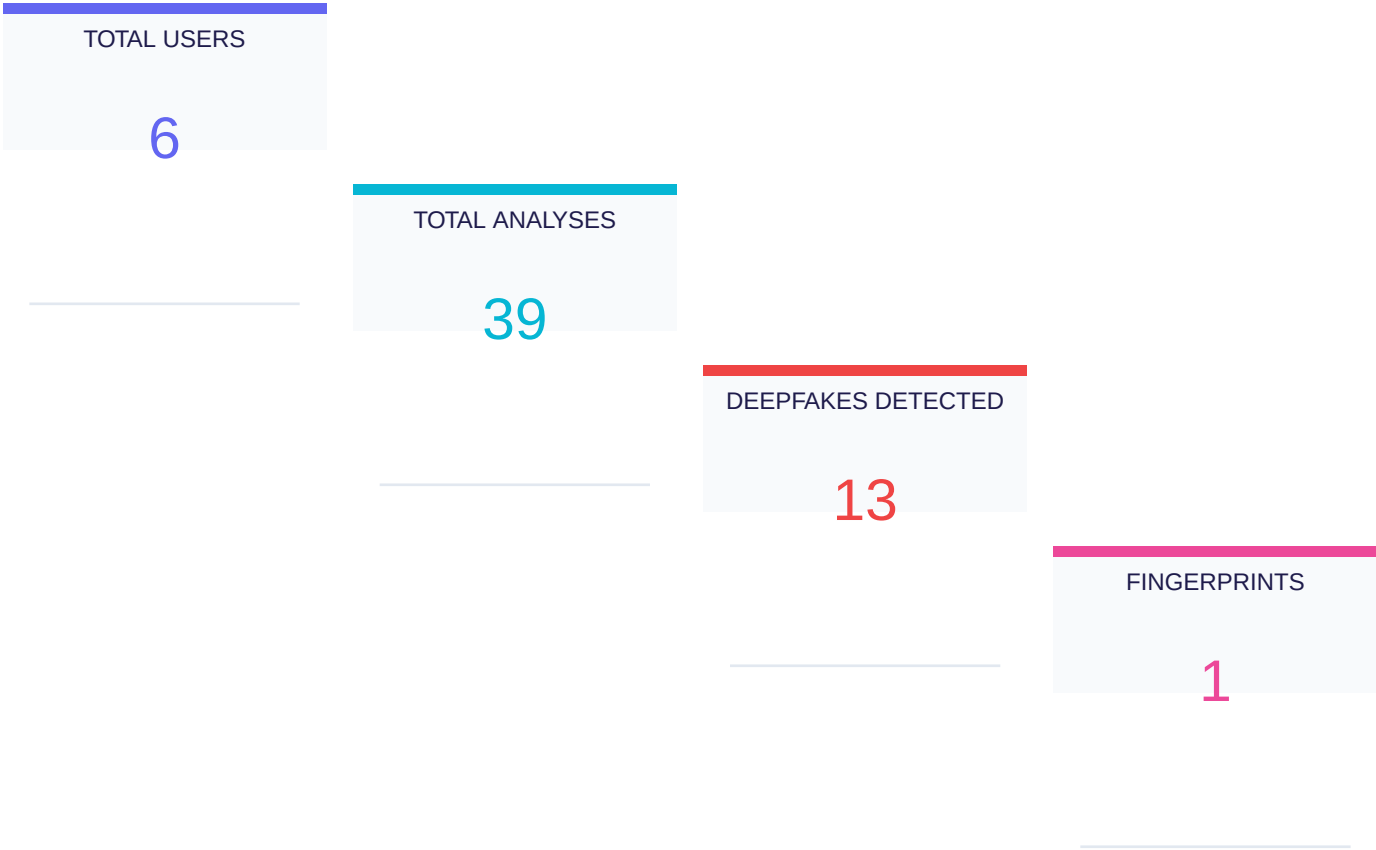
## PROJECT STATUS REPORT

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### Executive Summary

Naari Shourya (She Shield) is a comprehensive women safety platform that combines advanced AI-powered deepfake detection with reverse image search capabilities. The system enables users to analyze images for AI manipulation, search for similar images in a personal fingerprint database, and securely report incidents.

### Platform Statistics



### Core Features Implemented

■ AI-Powered Deepfake Detection

■ Active

■ Reverse Image Search

■ URL Analysis

■ Face Recognition & Fingerprinting

■ Secure Case Management

■ Emergency Alert System

Multi-layer analysis including GAN fingerprint, face warping, lighting consistency, and metadata analysis

■ Active Perceptual hash (pHash) based similarity matching with user fingerprint database

■ Active Direct URL scanning capability for images hosted externally

■ Active Personal photo registry for comparison against analyzed images

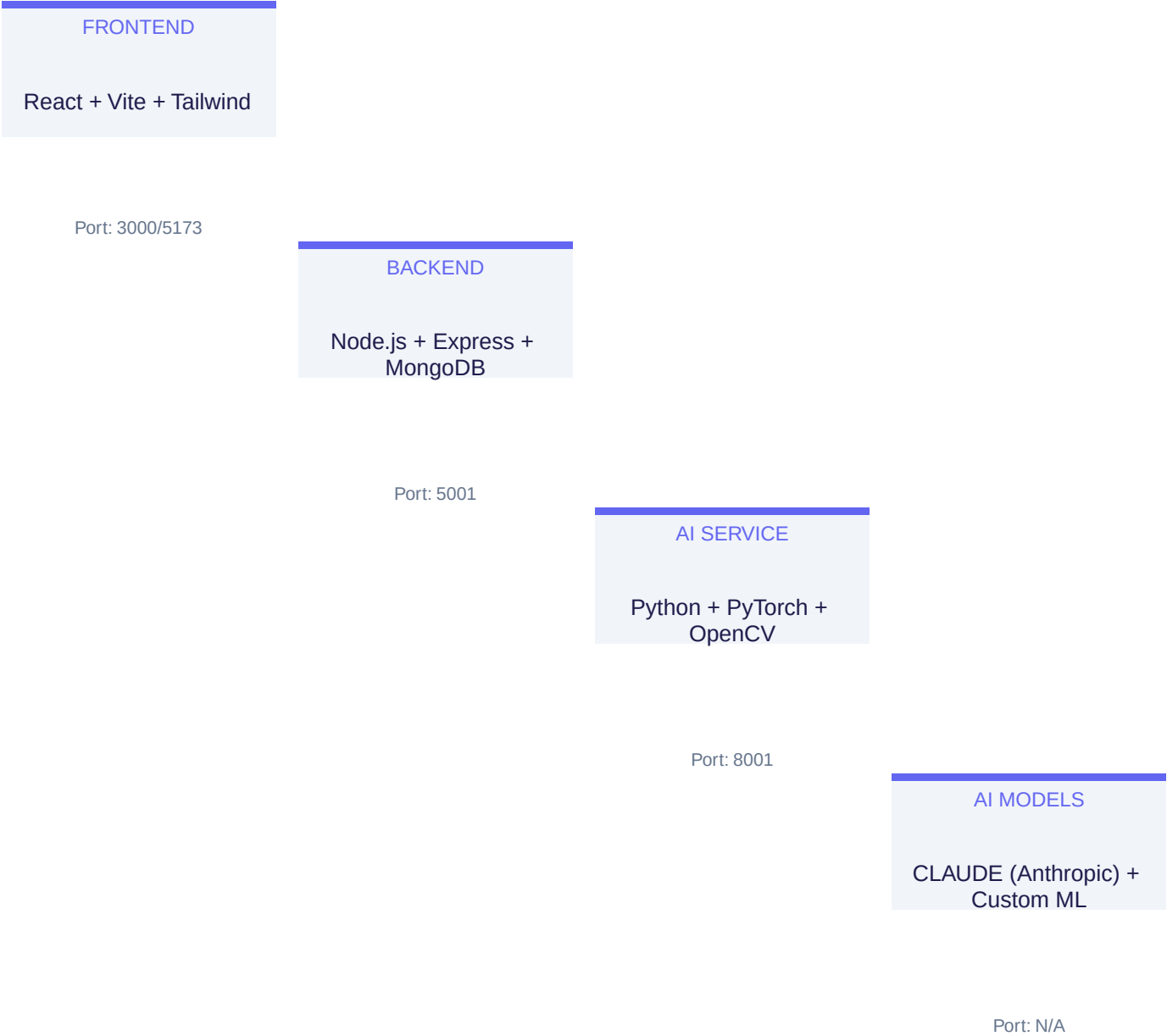
■ Active Anonymous and confidential reporting modes with case tracking

■ Active Quick SOS with trusted contacts and location sharing capabilities

Security &

- File Magic Byte Validation - Validates actual file content vs extension
- Input Sanitization - XSS, SQL/NoSQL injection protection on all routes
- Rate Limiting - Upload (20/hr), Auth (20/15min), Emergency alerts
- JWT Security - Secret validation with error throw (no insecure fallback)
- CORS Configuration - Whitelist based origin validation, returns 403 on rejection
- Malicious Content Detection - Hex dump analysis for embedded threats

Technical Architecture



## Improvements

- 1. MongoDB Schema Fix:  
Changed Analysis.result to Mixed type to preserve aiDetection, verdict, similarImages fields
- 2. AI False Positives:  
Added quality validation override - high Laplacian variance images get authenticity boost
- 3. Threshold Adjustment:  
Lowered deepfake threshold from <65 to <50 to reduce false positives
- 4. Dashboard Display:  
Fixed authenticity score showing 0% - now correctly reads from result.aiDetection.authenticityScore
- 5. CORS Response:  
Changed CORS rejection from 500 to 403 Forbidden
- 6. URL Scan Mode: Added URL toggle to Upload page with mode state and validation
- 7. Legacy Migration:  
Created migration script - fixed 24 of 35 legacy records with missing fields

## UI/UX Enhancements

- Complete AnalysisResult.jsx redesign with dark premium theme
- Animated score circle with gradient stroke and glow effect
- AI Generated badge with manipulation type display
- Red flags section for manipulated image indicators
- Upload.jsx file/URL mode toggle with validation
- Premium gradient backgrounds and glass-morphism effects
- Responsive design with mobile-friendly layouts

## API Endpoints Overview

|      |                                           |
|------|-------------------------------------------|
| POST | /api/auth/register<br>- User registration |
|      |                                           |

POST

POST

POST

POST

POST

GET